

Reinforcement Learning

Assignment Questions – Unit I & Unit II

1. Explain **Generative AI** and discuss its key concepts and core techniques such as **GANs, VAEs, Diffusion Models, and Transformer Architecture**.
2. Explain the **evolution of language models** from rule-based systems to modern Large Language Models (LLMs). Highlight the major developments at each stage.
3. Explain the **key features, capabilities, and common use cases of ChatGPT**. Also mention any two limitations of ChatGPT.
4. Discuss the major **ethical concerns associated with Generative AI** and explain the principles for its responsible usage.
5. Compare **Dynamic Programming, Monte Carlo methods, and Q-Learning** based on their requirement of an environment model, learning approach, and when the value update is performed.
6. Explain the **Q-Learning algorithm** with the Q-value, Q-table, learning rate, discount factor, and Q-Learning update rule.
7. Explain the **exploration vs. exploitation trade-off** in Q-Learning. Describe the **epsilon-greedy strategy** and explain how exploration decay helps during training.
8. Explain the **basic Agent–Environment interaction loop in Reinforcement Learning**. Describe the roles of **State, Action, Reward, Policy, and Episode** with a suitable example.
9. What is **Q-Learning**? Explain why it is called a **model-free and off-policy reinforcement learning algorithm**.
10. Explain the process of **training a Q-Learning agent for Tic-Tac-Toe**. Discuss the role of episodes, Q-table updates, exploration, and exploitation during training.
11. Explain the important **evaluation metrics for a Tic-Tac-Toe Q-Learning agent**. Discuss **win rate, draw rate, loss rate, and convergence speed**.
12. Explain the different **fine-tuning strategies for a Q-Learning agent**. Discuss the effect of **learning rate, exploration decay, reward shaping, and number of training episodes**.
13. Describe the complete **end-to-end workflow for the Classic RL Tic-Tac-Toe assignment**, starting from building the game environment to training, evaluation, and fine-tuning of the Q-Learning agent.